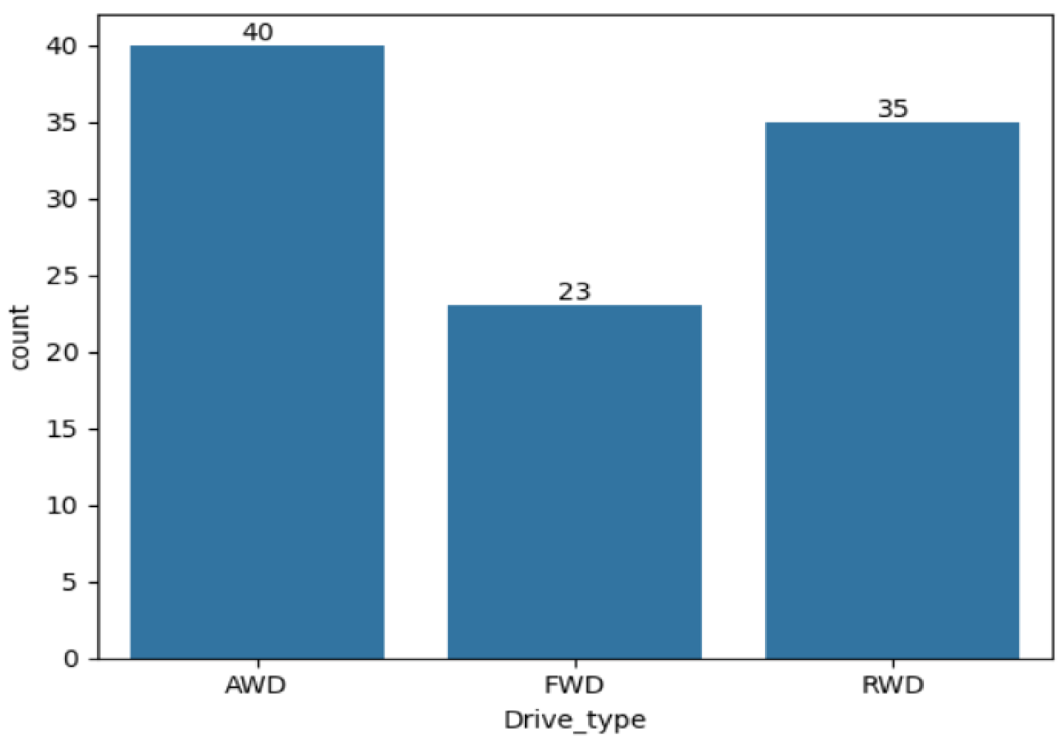
REPORT PROJECT ON EV MARKET OF CARS BY USING PYTHON NOTEBOOK

```
print(df.columns)
```

```
Index(['Brand', 'Model', 'Year', 'Variant', 'Price_usd',
       'Battery_capacity(kwh)', 'Range(miles)', 'Charging_speed(Kw)',
       'Horsepower', 'Torque(nm)', 'Drive_type', 'Seating_capacity',
       'Body_type', 'Market_segment', 'Annual_sales_units', 'Customer_rating',
       'Warranty_years', 'Price(INR)'],
      dtype='object')
```

```
ax =sns.countplot(x="Drive_type",data=df)

for bars in ax.containers:
    ax.bar_label(bars)
```



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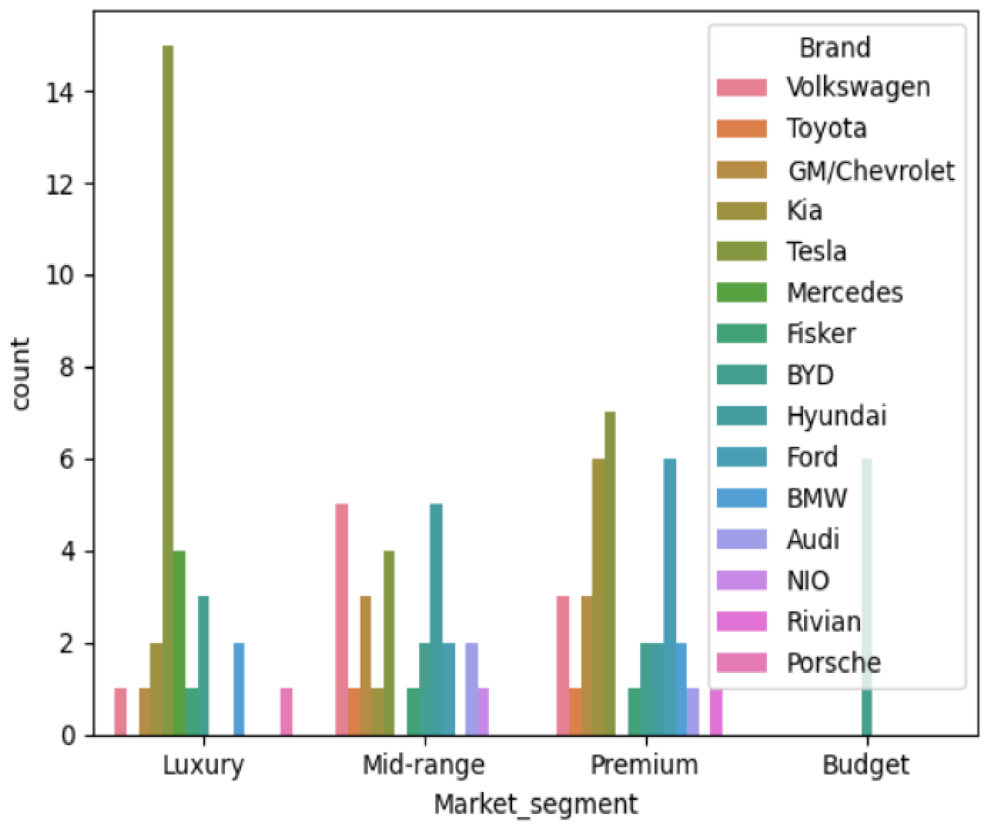
```
df .groupby(["Drive_type"],as_index=False)["Price(INR)"].sum().sort_values(by="Price(INR)", ascending=False)
```

	Drive_type	Price(INR)
0	AWD	3.288929e+08
2	RWD	3.033886e+08
1	FWD	1.473986e+08

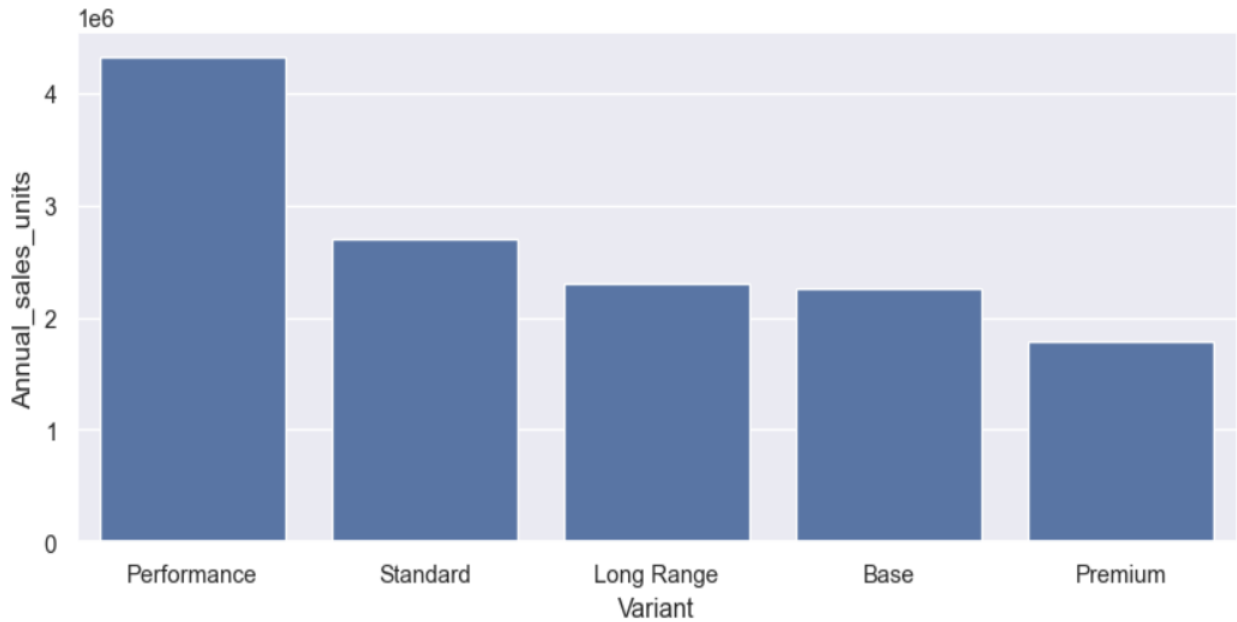
```
sns.countplot(data=df,x="Market_segment",hue="Brand")
```

```
<Axes: xlabel='Market_segment', ylabel='count'>
```

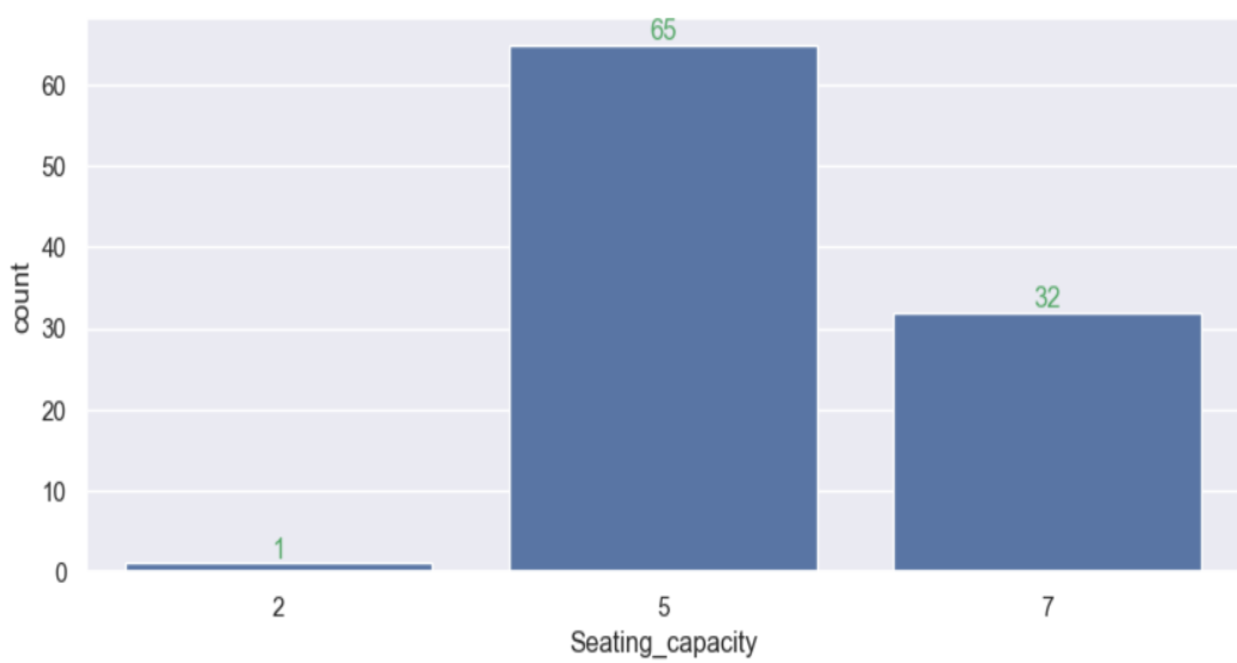


```
Sales_variant=df.groupby(["Variant"],as_index=False)["Annual_sales_units"].sum().sort_values
    (by="Annual_sales_units",ascending=False).head(10)
sns.set(rc={"figure.figsize":(10,4)})
sns.barplot(data=Sales_variant,x="Variant",y="Annual_sales_units")
```

```
<Axes: xlabel='Variant', ylabel='Annual_sales_units'>
```



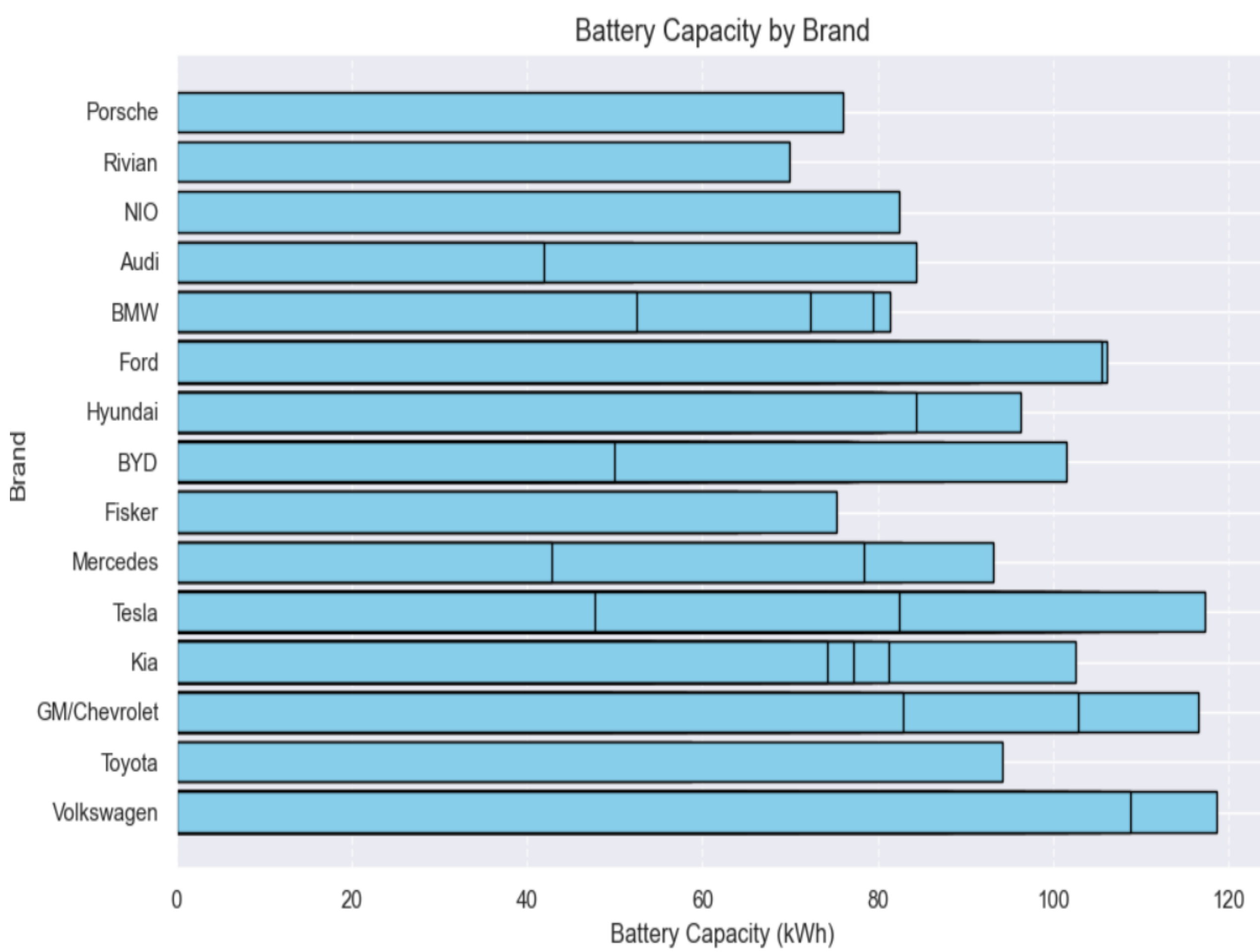

```
ax =sns.countplot(x="Seating_capacity",data=df)
for bars in ax.containers:
     ax.bar_label(bars,color="g")
```



REPORT PROJECT ON EV MARKET OF CARS BY USING PYTHON NOTEBOOK

```python
import matplotlib.pyplot as plt
import pandas as pd
data = pd.read_csv(r"C:\Users\harsh\OneDrive\Desktop\ev_market_2026.csv")
df = pd.DataFrame(data)

plt.figure(figsize=(10, 6))
plt.barh(df["Brand"], df["Battery_capacity(kwh)"], color='skyblue', edgecolor='black')
plt.title('Battery Capacity by Brand', fontsize=14)
plt.xlabel('Battery Capacity (kWh)', fontsize=12)
plt.ylabel('Brand', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



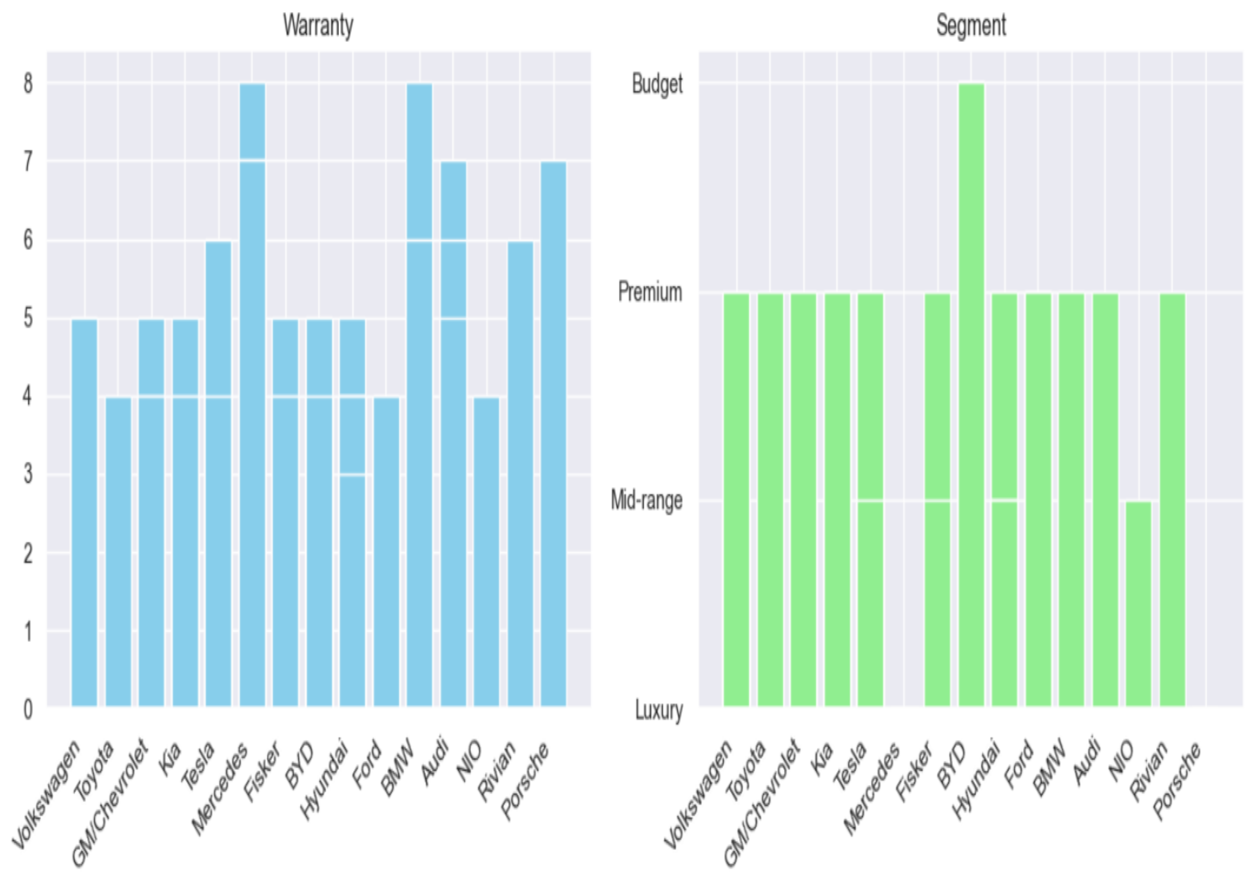
REPORT PROJECT ON EV MARKET OF CARS BY USING PYTHON NOTEBOOK

```
import matplotlib.pyplot as plt
import pandas as pd

data =pd.read_csv(r"C:\Users\harsh\OneDrive\Desktop\ev_market_2026.csv")
df = pd.DataFrame(data)

plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.bar(df['Brand'], df['Warranty_years'], color='skyblue')
plt.title('Warranty')
plt.xticks(rotation=45, ha='right')

plt.subplot(1, 2, 2)
plt.bar(df['Brand'], df['Market_segment'], color='lightgreen')
plt.title('Segment')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



REPORT PROJECT ON EV MARKET OF CARS BY USING PYTHON NOTEBOOK

```
import matplotlib.pyplot as plt

df_sorted = df.sort_values(by='Horsepower').reset_index(drop=True)
plt.figure(figsize=(12, 6))
plt.fill_between(df_sorted.index, df_sorted['Torque(nm)'], color='skyblue', alpha=0.5, label='Torque (nm)')
plt.plot(df_sorted.index, df_sorted['Torque(nm)'], color='deepskyblue', linewidth=2)

plt.fill_between(df_sorted.index, df_sorted['Horsepower'], color='lightgreen', alpha=0.5, label='Horsepower')
plt.plot(df_sorted.index, df_sorted['Horsepower'], color='forestgreen', linewidth=2)

plt.title('Horsepower vs. Torque Magnitude across EVs', fontsize=14)
plt.xlabel('Vehicles (Sorted from Lowest to Highest Horsepower)', fontsize=12)
plt.ylabel('Metric Value', fontsize=12)
plt.legend(loc='upper left')
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```